

- 2 The red squirrel is a tree-living mammal native to woodlands in the United Kingdom. The population and distribution of the red squirrel have been decreasing since the introduction of a North American species of grey squirrel. The grey squirrel occupies an ecological niche that overlaps with that of the red squirrel, but the grey squirrel is a better competitor.

One cause of the decline in red squirrels is the clearance of woodland for development. New areas of woodland have been planted, but the species of tree planted have changed over time.

Table 2.1 shows the changes in species composition of woodland in England and Scotland from 1980 to 1990 and the suitability of each species as habitats for red and grey squirrels.

Table 2.1

tree species		suitability of tree species as a habitat for squirrels	percentage change in species composition	
			England	Scotland
narrow-leaved trees	Sitka spruce	poor for both species	+14	+61
	Scots pine	poor for both species	−5	+8
	Norway spruce	good for both species	−21	−28
	Douglas fir	good for both species	+3	+1
	European larch	good for red squirrels	−34	−39
	Japanese larch	good for red squirrels	−3	+21
broad-leaved trees		good for grey squirrels	+36	+68

- (a) (i) Use the information in Table 2.1 to summarise the effects on habitat for these species of squirrel as a result of the changes in the tree species composition in woodland.

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- (ii) Suggest why the populations of red squirrels in England and Scotland have decreased.

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- (b) Squirrel parapoxvirus (SPPV) infects both red squirrels and grey squirrels. Grey squirrels develop immunity to SPPV, but red squirrels are killed by this virus.

One method used to conserve red squirrels on an island involved culling all the grey squirrels.

During the cull, research was carried out to find the percentage of grey squirrels that were immune to SPPV. When the cull started the island had an estimated population of 40 red squirrels and 3 000 to 4 000 grey squirrels.

The research involved several steps.

- Traps were placed at a density of 1 per 0.01 km² in broad-leaved woodland and 2 per 0.01 km² in narrow-leaved woodland.
- Humane steel traps, all of the same size, were used. Non-target animals caught were released, any grey squirrel was killed.
- Blood samples were taken from grey squirrels trapped between January and August at randomly selected habitats.
- A standard test for antibodies to SPPV was used to screen the blood samples.
- A control sample of grey squirrels from the mainland was tested in 2010.
- The percentage testing positive for antibodies to SPPV was calculated for each sample.

Table 2.2 shows the results of this research.

Table 2.2

year	number of grey squirrels tested	percentage positive for SPPV antibodies
1999	23	52
2002	20	75
2003	99	40
2007	54	13
2010	28	4
2010 (control)	70	67

- (i) State two variables that were controlled during the research.

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- (ii) The researchers concluded that culling could help to reduce the spread of SPPV.

State how the evidence in Table 2.2 supports this conclusion.

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Culling of grey squirrels on the island successfully removed all of them by 2013. Subsequently the population of red squirrels has increased.

Culling of grey squirrels has also occurred on the mainland in areas occupied by both species. Within a few months of the end of a cull, the grey squirrel population returns to pre-cull level and the red squirrel population is unchanged.

- (iii) Suggest why the cull on the mainland is less effective at removing grey squirrels than the cull on the island.

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[Total: 9]